

INTRODUCTION

1.1 Invitation

The study of physics since its modern formulation in the times of Newton and Galileo has found ever greater explanatory power in finding unifying principles that relate seemingly disparate phenomena. While this insight is not novel, the discovery of *dualities* between physical theories is a more recent take on the paradigm of unification, which has spurred several lines of inquiry in theoretical physics.

One such duality that arose from the study of amplitudes is **color-kinematics duality**, also known as **BCJ duality**¹ [1]. BCJ duality shows up as a relationship between amplitudes in Yang-Mills theories and gravitational theories — in a Feynman diagram expansion of Yang-Mills amplitudes, performing a term-by-term replacement of factors that carry all the color information by a kinematic factor results in a gravitational amplitude with the same external kinematics.

For instance, at tree level, a four-point scattering amplitude in pure Yang-Mills theory can be written as [1]

$$\mathcal{A}_4^{\text{tree}} = g^2 \left(\frac{n_s c_s}{s} + \frac{n_t c_t}{t} + \frac{n_u c_u}{u} \right), \quad (1.1)$$

where the three terms represent contributions from the s -, t -, and u -channel diagrams respectively.² The denominators s, t, u are the usual Mandelstam variables, $s = (p_1 + p_2)^2, t = (p_1 + p_4)^2, u = (p_1 + p_3)^2$, where p_i are the external gluon momenta. The terms in the numerator are split into *color factors* c_i , which depend only on the Yang-Mills symmetry group, and *kinematic factors* n_i , which depend only on external momenta and polarizations.

¹The two names are used interchangeably throughout this dissertation.

²In the usual Feynman diagram expansion, there is another diagram from the four-point vertex of a non-Abelian Yang-Mills theory. It turns out that the fourth diagram can be written as the sum of three terms, which can be absorbed appropriately into the three terms above.

in a way that does not obviously follow from the Lagrangians themselves. While our interest in the double copy paradigm was motivated by probing the extent of the double copy proposal and hopefully learning something fundamental about the nature of gravitational theories, the double copy has also ~~given rise to applications to~~ found applications in other fields of physics [4].

1.2 From Quantum Amplitudes to Classical Physics

The discovery of a double copy structure in the amplitudes computed in gauge and gravity theories naturally leads to a similar structure in their respective classical limits. Since the Feynman diagram series with an increasing number of loops is effectively an expansion in increasing powers of $\hbar$, taking the classical limit of an amplitude generally corresponds to keeping only the tree-level terms and discarding all diagrams with loops. See the recent review [5] for a comprehensive description of how the classical double copy emerges from the double copy of amplitudes.

The study of the classical double copy via amplitudes expanded beyond the amplitudes perspective and took on a life of its own after a seminal paper [6] that described how a certain class of spacetimes (Kerr-Schild spacetimes) could be constructed *exactly* from solutions in the corresponding electromagnetic and scalar theories. Kerr-Schild solutions are distinguished by the ability of the metric to be written as

$$g_{\mu\nu} = \eta_{\mu\nu} + \phi k_\mu k_\nu, \quad (1.6)$$

where k_μ is a null vector field tangent everywhere to affinely parametrized geodesics, ϕ is a scalar function, and $\eta_{\mu\nu}$ is the Minkowski metric. The authors of [6] showed that the corresponding single and zeroth copies are,

$$A_\mu = \phi k_\mu, \quad \Phi = \phi, \quad (1.7)$$

which satisfy equations of motion (for a $U(1)$ gauge field and a massless scalar field) when considered to be fields on Minkowski spacetime with metric $\eta_{\mu\nu}$. Since this proposal of the *Kerr-Schild double copy*, there have been proposals for other formulations that extend the classical double copy in various ways. The most notable of these for our purposes is the *Weyl double copy* [7, 8].

1.3.2 Horizons

A rich area of inquiry within the double copy paradigm ~~is~~ has been building out the double copy dictionary beyond simply calculating the gravitational, gauge, and scalar fields themselves. However, a question remained: how do black hole horizons emerge from the double copy structure of black hole spacetimes? If the double copy proposal is to be taken seriously, in that gravitational physics is entirely constructible from the same building blocks as the physics of a gauge field (plus an associated biadjoint scalar), there must be a way to identify the presence of a horizon from the single and zeroth copy theories.

For instance, the single copy of the Schwarzschild black hole is the Coulomb potential of a static point charge, and the zeroth copy (scalar field) is proportional to $1/r$. From these fields, it is not immediately obvious how the double copy's horizon at ~~$r = GM$~~ $r = 2GM$ could be diagnosed. In particular, the gauge potential and the scalar field are smooth everywhere except $r = 0$.

We explore this question in chapter 3, where we give a proposal for finding trapped surfaces in the gravitational solution using quantities that can be computed using the gauge field, the associated scalar, and the supposition of a Kerr-Schild solution that is their double copy. The relevant quantities involve both the gauge field and the scalar field together, but they are not invariant under gauge transformations of the gauge potential.

1.3.3 The Penrose limit

So far, we relied on double copy proposals for classical gravity that were both *exact*, and *local* in position space. These methods are known to work for special classes of spacetimes, one of which (Kerr-Schild metrics) we have already mentioned. The Weyl double copy in its exact form requires the spacetime to be *algebraically special*, which is a condition on the *principal null directions* of the Weyl tensor. More precise statements about algebraic speciality are made in [section 2.2.2.3](#).

The pursuit of ever more general examples of the double copy remains a fruitful endeavor. However, recent work [9] showed that the two formalisms we have been working with (Kerr-Schild and

Weyl) are expected to fail for algebraically general spacetimes. More than a failure of exactness, the authors of [9] set out to answer why the classical double copy was local in the fields anyway — the amplitudes double copy seems to imply that the double copy should be local in momentum space, not in position space. It turns out that the existence of a local double copy relies strongly on algebraic speciality. ■

It seems then that there is more to be gained from a perturbative approach. The perturbative classical double copy arising from the amplitudes paradigm is naturally expressed as a perturbation about Minkowski spacetime. Now, with the knowledge of exact classical double copies, we had the opportunity to deal with expansions about other background spacetimes. One such approach was taken by [10], where the Weyl spinor was expanded around null infinity in an asymptotically flat spacetime. The Weyl spinor is algebraically special to leading orders in $1/r$, which allows for application of the Weyl double copy near null infinity.

Our motivation was to pursue a related line of inquiry: what happens if, instead of expanding about a null boundary, we expand the Weyl spinor around a null geodesic in the bulk? The limiting spacetime (the *Penrose limit*) at any null geodesic in any spacetime was shown to be a plane wave by Penrose [11], which is an algebraically special spacetime.

Chapters 4 and 5 lay out our investigations in expanding about null geodesics. Chapter 4 is specifically about whether the process of taking the Penrose limit commutes with the double copy map, which we expected on physical grounds but nevertheless had to explicitly check. Chapter 5 then examines how the double copy structure of a spacetime decomposes into various terms in an expansion about the null geodesic. This allowed us to examine at what order in the Penrose expansion the existence of a local double copy breaks down for generic spacetimes.

References

- [1] Z. Bern, J. J. M. Carrasco, and H. Johansson, “New relations for gauge-theory amplitudes”, *Phys. Rev. D* **78**, 085011 (2008).

THE PENROSE LIMIT OF THE WEYL DOUBLE COPY

The contents of this chapter first appeared in Classical and Quantum Gravity [1].

We embed the Penrose limit into the Weyl classical double copy. Thereby, we provide a lift of the double copy properties of plane wave spacetimes into black hole geometries and we open a novel avenue towards taking the classical double copy beyond statements about algebraically special backgrounds. In particular, the Penrose limit, viewed as the leading order Fermi coordinate expansion around a null geodesic, complements approaches leveraging asymptotic flatness such as the asymptotic Weyl double copy. Along the way, we show how our embedding of the Penrose limit within the Weyl double copy naturally fixes the functional ambiguity in the double copy for Petrov type N spacetimes. We also highlight the utility of a spinorial approach to the Penrose limit. In particular, we use this spinorial approach to derive a simple analytical expression for arbitrary Penrose limits of four-dimensional, vacuum type D spacetimes.

4.1 Introduction

Gravity and gauge theory are related in surprising ways. Among these connections is BCJ duality (or color-kinematics duality), which provides a map between scattering amplitudes of gravitational and Yang-Mills theories [2–4]. This map is usually termed a “double copy” map, since it involves replacing color factors in each term by a second copy of the respective kinematic factor.

On the one hand, the double copy presents a powerful technical tool — gravitational amplitudes can be expressed in terms of simpler gauge theory amplitudes, which allows for efficient computations when modeling gravitational wave observations of black hole binary systems [5–9]. On the other hand, the double copy relation appears to persist in UV-complete theories of quantum gravity, for instance manifesting as the KLT relations between open and closed string amplitudes [10]. The double copy is thus arguably of foundational interest.

4.3, in which we outline the general procedure for taking a Penrose limit of a gravitational solution, along with the gauge theory and scalar solutions that are associated to it via its double copy structure. We illustrate our approach in detail for the Schwarzschild and Kerr solutions in sections 4.4.1 and 4.4.2, which has a straightforward generalization to the generic vacuum type D (Plebanski-Demianski) metric in section 4.4.3. We also examine type N spacetimes in 4.4.4. Detailed computations for the examples can be found in appendix D.

Conventions

We will generally follow the conventions of Penrose and Rindler [\[penrose1984spinors, 55\]](#) [\[55, 56\]](#). An important exception is the Riemann curvature which differs by a sign from that reference. See also Appendix C.

4.2 Background

In this section, we review the classical double copy and the Penrose limit, which we will subsequently combine. We highlight the main elements we need later. For a broader overview of the double copy and further references to the literature see [3, 57]. A more complete introduction to the Penrose limit can be found in [58].

4.2.1 Classical Double Copy

The double copy of exact classical solutions was first discussed in [18] for Kerr-Schild solutions. These spacetimes possess a metric that can be written (in Kerr-Schild coordinates [59]) as

$$g_{\mu\nu} = \eta_{\mu\nu} + \phi k_\mu k_\nu, \tag{4.2}$$

4.2.2 Penrose Limit

The Penrose limit was originally introduced in [54] starting from a coordinate system adapted to a twist-free null geodesic congruence. The metric in such coordinates in d dimensions, by definition, takes the form

$$ds^2 = 2dUdV + D(x^\mu)dV^2 + B_i(x^\mu)dVdX^i - C_{ij}(x^\mu)dX^idX^j, \quad (4.9)$$

with $D(x^\mu)$, $B_i(x^\mu)$, and $C_{ij}(x^\mu)$ functions of all the coordinates $x^\mu = \{U, V, X^i\}$ and with indices i, j running over the $d - 2$ “transverse” directions ($i, j, \dots \in \{1, \dots, d - 2\}$). Note that $C_{ij}(x^\mu)$ is symmetric and positive definite. The null coordinate vector field ∂_U is geodesic and it is in this sense that the coordinates are “adapted” to the null geodesic congruence generated by ∂_U .

The requirement that the geodesic congruence is twist-free arises as an integrability condition for it to be written as a coordinate vector field. On the other hand, any geodesic can locally be embedded in a twist-free geodesic congruence. More precisely, given any reference geodesic γ , a coordinate system can be found in a neighborhood of a portion of γ free of conjugate points²⁰ such that the metric takes the form (4.9) with (the conjugate point-free portion of) γ corresponding to the geodesic at $X^i = V = 0$ [penrose1972techniques][70].

Having constructed the adapted coordinates $\{U, V, X^i\}$ associated to γ , the Penrose limit amounts to the leading order of an expansion in $\epsilon \rightarrow 0$ where $\{U, V, X^i\} \sim \{\epsilon^0, \epsilon^2, \epsilon^1\}$. One way to interpret the scalings is by considering the combination of an ordinary expansion around a point but simultaneously boosting along the null geodesic direction $\{U, V, X^i\} \sim \epsilon \times \{\epsilon^{-1}, \epsilon, 1\}$. With these scalings, on the assumption that $D(x^\mu)$, $B_i(x^\mu)$, and $C_{ij}(x^\mu)$ are smooth,²¹ the metric scales as $ds^2 = O(\epsilon^2)$ and the Penrose limit line element ds_γ^2 is thus given by only keeping the terms in (4.9)

²⁰Conjugate points (or focal points) are a pair of points such that there exists a Jacobi vector field (or equivalently here: a non-trivial solution to the geodesic deviation equation) vanishing at both points. Conjugate points are sometimes referred to as caustics but, more commonly, caustics are the geometric locus of conjugate points for a family of geodesics [69].

²¹The smoothness assumption is made by [54] but can be somewhat relaxed. We will not attempt a precise formulation of weaker assumptions here but ideally shockwaves and other weak solutions to the Einstein equations would be included where possible.

From the form of the Weyl spinor (4.123) and the spinor prescription to take the Penrose limit (4.28), we find

$$ds_\gamma^2 = 2dudv - \frac{3K^2}{M^4} (M^5 \zeta^{-5} z^2 + M^5 \bar{\zeta}^{-5} \bar{z}^2) du^2 - 2dzd\bar{z}, \quad (4.130)$$

with $\zeta(u) = \zeta|_\gamma = r(u) - iy(u)$, and where $r(u)$, $y(u)$ satisfy the geodesic equations (4.126). In terms of real transverse coordinates, as in (4.22), the metric (4.130) becomes

$$ds_\gamma^2 = 2dudv - dx_1^2 - dx_2^2 - \frac{3K^2 M}{\Sigma^5} [(r^5 - 10r^3 y^2 + 5ry^4) (x_1^2 - x_2^2) - 2(5r^4 y - 10r^2 y^3 + y^5) x_1 x_2] du^2. \quad (4.131)$$

This result is consistent (up to rescaling of u and v) with equatorial and photon sphere results presented in [110, 111], and with the eigenvalues of the “wave profile” (H_{ab}) presented in [112] for general Petrov type D spacetimes as well as for Kerr in particular.³⁷

We will not describe here a parallel propagated null frame, derived from a dyad $\{\alpha, \beta\}$, nor the adapted coordinate approach as we did for the Schwarzschild black hole. We do wish to note, that after we computed (4.130) and (4.131), we became aware of the reference [116], where Penrose performs essentially the same calculation of (4.28) for vacuum Petrov type D spacetimes. However, Penrose doesn’t discuss it in the context of the Penrose limit or the geodesic deviation equations but rather as representing the “null datum” for the characteristic initial value problem [117]. Along the same lines, it was known (e.g. [\[penrose1984spinors\]](#)[\[56\]](#), Section 5.11) that (4.28) is a way to compute the “astigmatic”, $\Psi(u)$, and “anastigmatic”, $\Phi(u)$, part of the geodesic deviation but we are unaware of other sources than [76] emphasizing the simplicity of this approach, or actually using it to take a Penrose limit.

³⁷In fact, [112] may be considered to give a full alternative derivation for (4.130) when, in the right places, expressions are kept complex and the transverse directions are interpreted in terms of z and $\bar{z}$; see specifically their (7.46). However, this does not seem to have been what [112] had in mind.

Therefore, the Weyl double copy for type N spacetimes can be stated as

$$\Psi = \frac{f^2}{S}. \quad (4.149)$$

A condition on S for the compatibility of the Bianchi identity for Ψ with the Maxwell equation for f , on the curved background, for general type N spacetimes was established in [39]. The equations of motion for the field strength associated to f and the scalar S on the flat spacetime instead were established on a case by case basis. For the “twisted” type N metrics, where the principal null geodesic congruence has a non-vanishing twist, the flat space equations of motion were found not to be satisfied [39].

Focusing then on the non-twisting spacetimes with single copies on flat space, these type N metrics can naturally be given in terms of an adapted coordinate system⁴⁰ (of the form (4.9)) to the principal null geodesic congruence. A null affine time v along the principal null geodesic congruence is then used as a coordinate time and the principal null spinor can be chosen such that $(\partial_v)^{AA'} = \iota^A \bar{\iota}^{A'}$. Fixing the principal null spinor, it was shown in [39] that the functional ambiguity of the Weyl double copy can not in general be fixed by choices like (4.61) or (4.60).

On the other hand, the choice of principal null spinor is not unique. If we only impose that o^A is such that (4.147) holds, ι_A can be rescaled by an arbitrary scalar function. We will call this rescaling a Geroch-Held-Penrose (GHP) transformation [119], after the GHP, or compacted spin-coefficient, formalism in which one works covariantly with respect to such rescalings **[penrose1984spinors]** ■ [\[56\]](#). We will say a quantity X has boost-weight b and spin-weight s if under $o^A \rightarrow \lambda e^{i\varphi} o^A$ and $\iota^A \rightarrow \lambda^{-1} e^{-i\varphi} \iota^A$, for real λ and φ , it transforms as $X \rightarrow \lambda^{2b} e^{2is\varphi} X$. The zeroth copy scalar S , the field strength spinor f_{AB} , and the Weyl spinor Ψ_{ABCD} are invariant under GHP transformations but the (spacetime) scalars f and Ψ have boost-and spin weight 1 and 2 respectively. Moreover, the

⁴⁰In the example of the plane wave, the Brinkmann coordinates (4.11) are an adapted coordinate system for the principal, null geodesic congruence ∂_v . Hence, we avoid using the notation of (4.9) for adapted coordinate systems in which we would have $(U, V, X_1, X_2) = (v, u, x_1, x_2)$.

EXPANDING ON THE DOUBLE COPY IN NULL FERMI COORDINATES

The contents of this chapter first appeared in Classical and Quantum Gravity [1].

We propose a Weyl classical double copy for a Fermi normal coordinate expansion around null geodesics. To leading order in this “Penrose expansion”, we ~~re-cover~~ recover a previously proposed double copy of the Penrose limit. For spacetimes with an exact double copy, this Penrose limit double copy is extended to all orders. For ~~space-times~~ spacetimes without such a double copy, generic obstructions appear at second subleading order beyond the Penrose limit. We thus argue that for ~~any~~ any spacetime, near ~~any~~ any null geodesic there is a classical double copy structure at least up to first subleading order in the Penrose expansion. Finally, we point out a ~~difficulty~~ difficulty in identifying an appropriate flat space to extend our results to the Kerr-Schild double copy, related to the generic incompatibility between Kerr-Schild and ~~Penrose-Güven~~ Penrose-Güven gauge.

5.1 Introduction

The double copy has been a driving force in the study and calculation of amplitudes since the observation by [2] of a “color-kinematics duality”; expressing how relations between gauge theory amplitudes due to the color structure can be mirrored in their kinematical pieces. In addition to allowing the calculation of gravitational amplitudes using tools developed for calculating Yang-Mills amplitudes [3–7], the discovery of the double copy also led to a deeper examination of shared structure between various gauge and gravity theories that was previously unknown. See [8–10] for recent reviews.

A deceptively simple question is whether or not the double copy structure of amplitudes can also be seen directly in the classical limit of field theories. Early attempts to relate the amplitudes double copy to classical solutions are summarized in chapter 8 of [8]. The first *exact* formulation of the double copy for classical solutions was described in [11] for Kerr-Schild spacetimes. The Weyl double

Chapter 6

CONCLUSION

In this dissertation I have explained how I, along with my co-authors, investigated various extensions of the classical double copy.

To summarize our results, we found an extension of the Weyl double copy, which is formulated in terms of spinors, to a large class of five dimensional spacetimes (the Kerr-NUT-(A)dS spacetimes) using an existing proposal on five dimensional spinor representations. We additionally showed that the double copy structure for Kerr-NUT-(A)dS is consistent with the double copy structure found using the Kerr-Schild double copy, as well as with existing literature that used the Kerr-Schild double copy for various special cases within the Kerr-NUT-(A)dS class.

Next, in trying to understand the emergence of black hole horizons in the double copy paradigm, we found a way to diagnose and locate apparent horizons in Kerr-Schild spacetimes. This involved noticing that for Kerr-Schild spacetimes, the expansions of null geodesic congruences that are tangent to the Kerr-Schild vector field can be rewritten using the single copy gauge field and the zeroth copy scalar. Thus, for ~~stationary~~ Kerr-Schild black holes that have horizons ~~generated by a~~ that are not generated by the Kerr-Schild vector field, in principle the location of their horizons can be found via a calculation that uses only single and zeroth copy objects.

This approach is not without its drawbacks; for one, the calculation that we end up doing in the single and zeroth copy does not respect the gauge freedom of the single copy. This does not by itself disprove our approach — the Kerr-Schild double copy is itself gauge dependent, and thus it is not a surprise that our proposal for finding horizons is not gauge invariant. However the gauge dependence does mean that it is impossible to physically interpret the quantities calculated if we were to restrict ourselves to a naive combination of the single and zeroth copy theories, since all observable quantities should be gauge invariant.

Lastly, we turned to the question of the double copy structure of generic spacetimes. Our approach was to start by restricting to a very special subset of spacetime ~~(, a null geodesic),~~ and examining the

double copy structure of spacetime around the null geodesic to various orders ~~of~~in the scaling parameter. The lowest order corresponds to the Penrose limit, where the spacetime metric (appropriately scaled) is that of a plane wave, whose double copy structure is fairly well understood. We found that the procedure of taking the Penrose limit and of expanding about it is consistent with the double copy. Moreover, we saw that for a generic spacetime, the existence of a local Weyl double copy encounters an obstruction at second order beyond the Penrose limit.

As much as possible, we will follow the conventions of Penrose and Rindler [\[penrose1984spinors\]](#) [\[227\]](#). On matters of curvature however, we will deviate from [\[penrose1984spinors\]](#) [\[227\]](#) by a sign. The reason is that the definition of [\[penrose1984spinors\]](#) [\[227\]](#) for the curvature is equivalent to

$$[\nabla_\alpha, \nabla_\beta]V^\delta = R^{(PR)}_{\alpha\beta\gamma}{}^\delta V^\gamma, \quad (\text{C.1})$$

while we use

$$[\nabla_\alpha, \nabla_\beta]V^\delta = R^\delta_{\gamma\alpha\beta}V^\gamma. \quad (\text{C.2})$$

Otherwise, the conventions of [\[penrose1984spinors\]](#) [\[227\]](#) imply that we work in a mostly minus signature $(+, -, -, -)$ and we take the anti-symmetric $\epsilon_{AB} = -\epsilon_{BA}$ to lower spinor indices “left” and raise “right”

$$\kappa^A \epsilon_{AB} = \kappa_B, \quad \kappa^A = \epsilon^{AB} \kappa_B. \quad (\text{C.3})$$

We shall refer to a spin frame or dyad as a pair $\{o^A, \iota^A\}$ such that

$$\epsilon_{AB} o^A \iota^B = 1. \quad (\text{C.4})$$

Further implications of these spinor conventions are discussed in detail in [\[penrose1984spinors\]](#) [\[227\]](#).

The decomposition of the Riemann tensor and anti-symmetric field strengths into spinor components plays an important role in the Weyl double copy and thus in the main text. For an anti-symmetric tensor field $F_{\alpha\beta}$, the main observation is that it can be decomposed into

$$F_{\alpha\beta} = f_{AB} \bar{\epsilon}_{A'B'} + \bar{f}_{A'B'} \epsilon_{AB}. \quad (\text{C.5})$$

Similarly, the Weyl tensor can be decomposed into

$$C_{\alpha\beta\gamma\delta} = \Psi_{ABCD} \bar{\epsilon}_{A'B'} \bar{\epsilon}_{C'D'} + \bar{\Psi}_{A'B'C'D'} \epsilon_{AB} \epsilon_{CD}. \quad (\text{C.6})$$

The relation between the decompositions (C.5) and (C.6) is key to the Weyl double copy which, given additionally the complex “zeroth copy” scalar S gives a Weyl tensor (C.6) in terms of a 2-form (C.5) as (4.4)

$$C_{\alpha\beta\gamma\delta}^{(\text{WeylDC})} = \frac{1}{S} f_{(AB} f_{CD)} \bar{\epsilon}_{A'B'} \bar{\epsilon}_{C'D'} + \frac{1}{S^*} \bar{f}_{(A'B'} \bar{f}_{C'D')} \epsilon_{AB} \epsilon_{CD}. \quad (\text{C.7})$$

Various curvature and connection components with respect to a null basis have been conventionally associated to particular notations in what is usually called the Newman-Penrose formalism. While we will not lean on this heavily, we next summarize these symbols as used in the main text.

C.1 Newman-Penrose Formalism

We now introduce our conventions for the Newman-Penrose formalism, which is mostly an exercise in decomposing spinors into the dyad $\{o^A, \iota^A\}$ and tensors into the Newman-Penrose frame $\{k^\mu, n^\mu, m^\mu, \bar{m}^\mu\}$.

Let us start with an anti-symmetric tensor field $F_{\alpha\beta}$, as decomposed into spinors in (C.5). The Newman-Penrose field strength scalars f_0, f_1 and f_2 with respect to a dyad $\{o^A, \iota^A\}$ can be defined in terms of the two-spinor formulation as

$$f_{AB} = f_2 o_A o_B - 2f_1 o_{(A} \iota_{B)} + f_0 \iota_A \iota_B. \quad (\text{C.8})$$

Using (C.5), this is expressed in terms of the tensor itself with respect to the associated null frame $\{k^\mu, n^\mu, m^\mu, \bar{m}^\mu\}$ as

$$F_{\alpha\beta} = 2f_2 k_{[\alpha} m_{\beta]} - 2f_1 (k_{[\alpha} n_{\beta]} - m_{[\alpha} \bar{m}_{\beta]}) + 2f_0 \bar{m}_{[\alpha} n_{\beta]} + cc, \quad (C.9)$$

where we will use “cc” to abbreviate the complex conjugate piece of the expression. A consequence of (C.9), that we will regularly use for calculations, is

$$f_2 = -F_{\alpha\beta} n^\alpha \bar{m}^\beta, \quad f_1 = -\frac{1}{2} F_{\alpha\beta} (n^\alpha k^\beta + m^\alpha \bar{m}^\beta), \quad f_0 = F_{\alpha\beta} k^\alpha m^\beta. \quad (C.10)$$

The analogous procedure for the Riemann tensor is somewhat more (algebraically) complicated but is reviewed in detail in Chapter 4 of [\[penrose1984spinors\]\[227\]](#). First, introduce the curvature spinors Ψ_{ABCD} and $\Phi_{ABC'D'}$ as well as their complex conjugates and the scalar Λ by the decomposition

$$-R_{\alpha\beta\gamma\delta} = (\Psi_{ABCD} + \Lambda (\epsilon_{AC}\epsilon_{BD} + \epsilon_{AD}\epsilon_{BC})) \bar{\epsilon}_{A'B'} \bar{\epsilon}_{C'D'} + \Phi_{ABC'D'} \bar{\epsilon}_{A'B'} \epsilon_{CD} + cc. \quad (C.11)$$

Now, the Weyl and Ricci scalars are again essentially just components of Ψ_{ABCD} and $\Phi_{ABC'D'}$ with respect to the dyad $\{o^A, \iota^A\}$, similar to (C.9) for the Maxwell scalars⁵³

$$\begin{aligned} \Psi_0 &= \Psi_{ABCD} o^A o^B o^C o^D = -C_{\alpha\beta\gamma\delta} k^\alpha m^\beta k^\gamma m^\delta, \\ \Psi_1 &= \Psi_{ABCD} o^A o^B o^C \iota^D = -C_{\alpha\beta\gamma\delta} k^\alpha n^\beta k^\gamma m^\delta, \\ \Psi_2 &= \Psi_{ABCD} o^A o^B \iota^C \iota^D = -C_{\alpha\beta\gamma\delta} k^\alpha m^\beta \bar{m}^\gamma n^\delta, \\ \Psi_3 &= \Psi_{ABCD} o^A \iota^B \iota^C \iota^D = -C_{\alpha\beta\gamma\delta} k^\alpha n^\beta \bar{m}^\gamma n^\delta, \\ \Psi_4 &= \Psi_{ABCD} \iota^A \iota^B \iota^C \iota^D = -C_{\alpha\beta\gamma\delta} n^\alpha \bar{m}^\beta n^\gamma \bar{m}^\delta, \end{aligned} \quad (C.12)$$

and

$$\Phi_{00} = \Phi_{ABC'D'} o^A o^B \bar{o}^{C'} \bar{o}^{D'} = \frac{1}{2} R_{\mu\nu} k^\mu k^\nu, \quad (C.13)$$

etc.

We do not enumerate all of the $\Phi_{ABC'D'}$ components as we do not need most of them explicitly. On the other hand, the component Φ_{00} is important as it is the only (potentially) non-vanishing component of the Ricci spinor in a plane wave spacetime, as discussed around (4.34) in the main text. Similarly, the only non-vanishing Weyl scalar, in our convention, of a type N spacetime is Ψ_0 .

Given that the Weyl curvature $C_{\alpha\beta\gamma\delta}$ is given by

$$-C_{\alpha\beta\gamma\delta} = \Psi_{ABCD} \bar{\epsilon}_{A'B'} \bar{\epsilon}_{C'D'} + \bar{\Psi}_{A'B'C'D'} \epsilon_{AB} \epsilon_{CD}, \quad (C.14)$$

let us check some of the equalities in (C.12). In particular, for the main text, it is important that the null Fermi coordinate approach to the Penrose limit (4.17) [57, 59]

$$H_{ab}(u) = \left(R_{\mu\nu\alpha\beta} E_a^\mu u^\nu E_b^\alpha u^\beta \right)_{|_\gamma}. \quad (C.15)$$

⁵³Let us emphasize again that these are identical to the definitions of [\[penrose1984spinors\]\[227\]](#), given that we introduce an extra minus sign to correct for the different curvature convention (see (4.11.9) and (4.11.10)). On the other hand, these differ in sign from say [233], where a mostly plus signature is used.

such that in particular

$$\begin{aligned}\nabla_\mu \kappa^A &= \nabla_\mu (\kappa^o o^A + \kappa^l \iota^A) \\ &= (\partial_\mu \kappa^o + \gamma_{\mu o}^o \kappa^o + \gamma_{\mu l}^o \kappa^l) o^A + (\partial_\mu \kappa^l + \gamma_{\mu o}^l \kappa^o + \gamma_{\mu l}^l \kappa^l) \iota^A.\end{aligned}\quad (\text{C.26})$$

One way to compute (C.25) is using the associated Newman-Penrose frame. Introducing first the Newman-Penrose spin coefficient symbols of [\[penrose1984spinors, 228\]](#)[\[227, 228\]](#), which give names to the individual components,

$$\begin{aligned}\kappa &= m^\alpha k^\mu \nabla_\mu k_\alpha, & \epsilon &= \frac{1}{2} (n^\alpha k^\mu \nabla_\mu k_\alpha + m^\alpha k^\mu \nabla_\mu \bar{m}_\alpha), & \gamma' &= \frac{1}{2} (k^\alpha k^\mu \nabla_\mu n_\alpha + \bar{m}^\alpha k^\mu \nabla_\mu m_\alpha), \\ \rho &= m^\alpha \bar{m}^\mu \nabla_\mu k_\alpha, & \alpha &= \frac{1}{2} (n^\alpha \bar{m}^\mu \nabla_\mu k_\alpha + m^\alpha \bar{m}^\mu \nabla_\mu \bar{m}_\alpha), & \beta' &= \frac{1}{2} (k^\alpha \bar{m}^\mu \nabla_\mu n_\alpha + \bar{m}^\alpha \bar{m}^\mu \nabla_\mu m_\alpha), \\ \sigma &= m^\alpha m^\mu \nabla_\mu k_\alpha, & \beta &= \frac{1}{2} (n^\alpha m^\mu \nabla_\mu k_\alpha + m^\alpha m^\mu \nabla_\mu \bar{m}_\alpha), & \alpha' &= \frac{1}{2} (k^\alpha m^\mu \nabla_\mu n_\alpha + \bar{m}^\alpha m^\mu \nabla_\mu m_\alpha), \\ \tau &= m^\alpha n^\mu \nabla_\mu k_\alpha, & \gamma &= \frac{1}{2} (n^\alpha n^\mu \nabla_\mu k_\alpha + m^\alpha n^\mu \nabla_\mu \bar{m}_\alpha), & \epsilon' &= \frac{1}{2} (k^\alpha n^\mu \nabla_\mu n_\alpha + \bar{m}^\alpha n^\mu \nabla_\mu m_\alpha), \\ \tau' &= \bar{m}^\alpha k^\mu \nabla_\mu n_\alpha, & \sigma' &= \bar{m}^\alpha \bar{m}^\mu \nabla_\mu n_\alpha, & \rho' &= \bar{m}^\alpha m^\mu \nabla_\mu n_\alpha, & \kappa' &= \bar{m}^\alpha n^\mu \nabla_\mu n_\alpha,\end{aligned}\quad (\text{C.27})$$

then

$$\begin{aligned}\gamma_{oo'o}^o &= \epsilon, & \gamma_{oo'o}^l &= -\kappa, & \gamma_{oo'l}^o &= -\tau', & \gamma_{oo'l}^l &= \gamma' \\ \gamma_{\iota o'o}^o &= \alpha, & \gamma_{\iota o'o}^l &= -\rho, & \gamma_{\iota o'l}^o &= -\sigma', & \gamma_{\iota o'l}^l &= \beta' \\ \gamma_{ol'o}^o &= \beta, & \gamma_{ol'o}^l &= -\sigma, & \gamma_{ol'l}^o &= -\rho', & \gamma_{ol'l}^l &= \alpha' \\ \gamma_{u'o}^o &= \gamma, & \gamma_{u'o}^l &= -\tau, & \gamma_{u'l}^o &= -\kappa', & \gamma_{u'l}^l &= \epsilon' .\end{aligned}\quad (\text{C.28})$$

Although we will not generally need the expressions (C.28) and symbols (C.27) in the main text, they are useful performing explicit calculations for the examples. Moreover, the expressions above quoted from [\[penrose1984spinors, 228\]](#)[\[227, 228\]](#) retain the possibility to work with an unnormalized dyad, while we always impose (C.4). Therefore

$$\epsilon = -\gamma', \quad \alpha = -\beta', \quad \beta = -\alpha', \quad \gamma = -\epsilon'. \quad (\text{C.29})$$

Several chapters of this dissertation are from previously published work. ~~Each such chapter has a declaration to that effect in the first line.~~ Chapter 2 is a reproduction of [83]. Chapter 3 is a reproduction of [84]. Chapter 4 is a reproduction of [82]. Chapter 5 is a reproduction of [81].

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